HR Workforce Insights & Employee Attrition Predictor Dashboard

PROJECT SUMMARY

A beginner-friendly Power BI portfolio project for exploring workforce composition, employee attrition patterns, and practical HR retention actions.

TOTAL RECORDS	ATTRITED RECORDS	CURRENT EMPLOYEE RECORDS	SAMPLE ATTRITION PROPORTION
200	39	161	19.5%

1. Project Objective

The project uses Power BI to turn employee records into interactive workforce insights. It explores where attrition is concentrated, how employee groups differ, and which questions HR should investigate before choosing retention interventions.

2. Tools and Dataset

- Microsoft Power BI Desktop for Power Query data cleaning, DAX measures, report visuals, and slicers.
- Excel / CSV dataset containing 200 synthetic employee records.
- Fields include department, job role, age, tenure, monthly income, overtime status, work mode, job satisfaction, work-life balance, attrition status, and illustrative risk category.

3. Dashboard Pages

Page	Purpose
1. Executive Overview	KPI cards summarizing headcount, current employee records, attrition count, sample attriti
2. Attrition Analysis	Attrition by department and job role, and breakdown by attrition status.
3. Workforce Insights	Employee composition and comparisons across workforce characteristics.
4. Retention Insights	Overtime, satisfaction, work-life balance, and illustrative risk categories to support further in

4. Sample Business Insights

The following figures were calculated from the original synthetic sample dataset. They may differ if rows or labels were changed during cleaning.

Department attrition

Sales and Finance have the highest attrition counts at 8 each. Sales has 8 of 30 records marked attrited (26.7%); Finance has 8 of 32 (25.0%).

Shorter tenure

Among records with 0 years at the company, 12 of 18 are marked attrited (66.7%). This suggests early-tenure retention deserves investigation in this sample.

Overtime

19 of 61 overtime records are marked attrited (31.1%), compared with 20 of 139 non-overtime records (14.4%). Investigate workload and scheduling; this is an association, not proof of causation.

Job satisfaction

Attrition proportions by satisfaction score: score 1 = 44.0% (11/25); score 2 = 21.6% (8/37); score 3 = 16.9% (13/77); score 4 = 11.5% (7/61). The lowest satisfaction group has the highest sample proportion.

Work-life balance

Score 1 = 22.2% (4/18); score 2 = 17.9% (7/39); score 3 = 19.1% (17/89); score 4 = 20.4% (11/54). The pattern is not consistently increasing or decreasing.

Job roles

Financial Analyst and Sales Manager each have 4 attrited records. Financial Analyst has 4 of 8 records marked attrited (50.0%); interpret carefully because the group is small.

Illustrative risk categories

High risk: 3/4 attrited; Medium risk: 25/89; Low risk: 11/107. These labels are rule-based examples, not validated model predictions.

5. Management Recommendations

- Investigate Sales, Finance, and Human Resources for potential workload, staffing, career-development, and employee-support issues.
- Strengthen onboarding, mentoring, and manager check-ins for employees in their early tenure.
- Review overtime patterns and staffing distribution where higher attrition proportions appear.
- Use employee feedback to investigate low satisfaction and improve recognition, communication, and career growth.
- Review recruitment, development, and succession plans for roles such as Financial Analyst and Sales Manager.
- Define a reporting period and track retention measures over time to evaluate whether interventions are effective.

6. Limitations and Interpretation

- The dataset is synthetic and does not describe a real employer.
- The 19.5% figure is a sample attrition proportion (39 of 200 records), not an annual attrition rate; exit dates and a defined reporting period are not available.
- Risk categories are illustrative and are not validated machine-learning predictions.
- Observed differences are associations and do not establish that overtime, satisfaction, or work-life balance causes attrition.

• Group sizes should be reviewed alongside percentages, especially for small job-role and risk groups.

7. Final Deliverables Checklist

- ■ Power BI report with four pages: Executive Overview, Attrition Analysis, Workforce Insights, and Retention Insights.
- ■ DAX measures, KPI cards, charts, summary matrix, and interactive slicers.
- ■ Project screenshots and this project summary PDF.
- ■ README and business insights documentation for portfolio publishing.

Prepared as a project summary for the HR Workforce Insights & Employee Attrition Predictor Dashboard.